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Large Scale Sampling Model and Optimization Algorithm Based On Dynamic Programming and Monte Carlo Simulation

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Abstract

The qualified rate of product parts to determine the excellence of the product, for large-scale parts sampling and inspection of the high cost, nowadays, large-scale parts sampling and testing methods are scarce, how to ensure that the qualified rate of parts and reduce the cost of sampling and inspection is the primary task. This paper proposes a sampling method for large-scale based on this. Firstly, this paper establishes a sequential sampling inspection method model based on the parts data of an electronic enterprise, secondly, establishes a cost-benefit model based on different sampling costs, and then introduces the method of dynamic programming on this basis to optimize the cost-benefit of sampling, and finally, conducts simulated sampling in the use of Monte Carlo simulation to obtain the sampling decision of the enterprise. This method overcomes the high-cost difficulties of large-scale sampling in reality and provides an effective method for sampling large totals.

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1. Introduction

The production process decision of the best-selling electronic products is often complex, and the sampling cost for such large-volume products is often huge, and most of the international sampling strategies are biased towards fixed and small samples, so this paper proposes a sampling model based on sequential sampling and sampling cost

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benefit optimization and simulation solution through Monte Carlo simulation to obtain the optimal sampling strategy, which provides the enterprise with a finished product production quality assurance and reduce the sampling cost.

In this paper, firstly, a model of sequential sampling inspection method is established based on the part data provided by the enterprise; secondly, a sampling cost-benefit model is established based on different sampling costs and other variables, and on this basis, a dynamic programming method is introduced to optimize the cost-benefit of sampling; lastly, simulated sampling is carried out by using Monte Carlo simulation to derive the sampling decision of the enterprise. The method overcomes the high-cost difficulties of large-scale sampling in reality and provides an effective method for large-scale sampling in enterprise manufacturing, which reduces the sampling cost and improves the sales revenue of the finished products while guaranteeing the qualified rate of the finished products.

2. Related work

Internationally, Miroslav V. et al.[1] proposed a technique based on an adaptive sequence sampling method, which significantly reduces the computational effort in engineering mechanics and applied mechanics by dynamically adjusting the sampling strategy. Scagliarini M. et al.[2][3] proposed a sequential test based on process capability index, which optimizes the statistical test process and improves the accuracy of the assessment of process capability during the sampling process, and is applicable to the field of quality control. Zylberberg A. et al.[4][5] based on the sequence sampling model, a value construction process is proposed to explain the phenomenon of sequence dependence in decision making. The explanatory power of the model for value judgments and decision-making behavior in the field of cognitive science is verified. Fomichev M V. et al.[6][7] investigated the complexity of sequence sampling methods and proposed improvement strategies to solve the efficiency problems in industrial mathematics and application scenarios. The computational complexity of the sequence sampling algorithm was optimized by mathematical analysis. Fan M et al. [8] proposed a method for mass sampling which reduces the sampling cost and improves the efficiency of sampling and the reliability of detection. Bloise G et al. [9] proposed an approximation method for dynamic programming for solving optimization problems with unbounded returns, which reduces the computational complexity of the dynamic programming problem through approximation techniques. Priorelli M et al.[10][11] proposed a dynamic programming method based on hierarchical active reasoning, which improves the computational efficiency of dynamic programming in complex environments. Psarommatis F et al.[12] constructed a cost-benefit based model for evaluating the sustainability of product reuse and recycling in circular remanufacturing, which incorporates both economic and environmental factors to provide a quantitative basis for decision making in the circular economy. Lovasz Z et al.[13] proposed the use of a geometric analysis method based on Monte Carlo simulation to investigate the effect of geometric parameters in quadrupole impedance spectroscopy testing. Dario C F et al.[14] investigated the long-term extreme jitter response of long-span bridges based on Monte Carlo simulation method of importance sampling, and proposed a simulation method for efficiently assessing the probability of extreme response of bridges. Huang T et al.[15][16] improved the Bayesian model averaging method, and improved the convergence speed and accuracy of Monte Carlo simulation by introducing the adaptive sampling technique.

3. Methods

3.1 Sequential Sampling Model Based on Parts Data of Electronic Enterprises

1) Electronic Enterprise Parts Data

Tables 1 and 2 show data on parts and defect rates.

Table 1. Parts Failure Rate and Cost Data

Part 1			Part 2		
Defective rate	Prices	Testing costs	Defective rate	Prices	Testing costs
0.1	4	2	0.1	18	3
0.2	4	2	0.2	18	3
0.1	4	2	0.1	18	3
0.2	4	1	0.2	18	1
0.1	4	8	0.2	18	1
0.05	4	2	0.05	18	3

Table 2. Cost and selling price of finished goods and cost of recovery of Unqualified products

Finished goods			Unqualified products	
Defective rate	Testing costs	Prices	Exchange costs	Disassembly costs
0.1	3	50	6	5
0.2	3	50	6	5
0.1	3	50	30	5
0.2	2	50	30	5
0.1	2	50	10	5
0.05	3	50	10	40

2) Sequential Sampling Mode

Sequential sampling is a dynamic sampling method that determines whether to continue sampling by gradually increasing the sample size and analyzing the results based on the current data. Unlike traditional fixed sample size sampling, Sequential Sampling allows the results of an experiment to be evaluated in real time during the data collection process to determine whether to stop sampling or continue to collect more data. The basic principle of sequential sampling is to update the decision rules during the course of the experiment, and gradually decide whether or not to continue data collection based on the results of statistical analysis of the existing data. This method is able to minimize the sample size within a given confidence level and margin of error, thereby increasing sampling efficiency and reducing sampling costs.

Assume that the actual defective rate of a batch of parts is p , and the acceptance rate is $1 - p$. For each part (Part 1 or Part 2), the number of defective items X follows a "0-1" distribution:

$$P(X = 1) = p, \text{ or } P(X = 0) = 1 - p. \quad (1)$$

If the batch quality is poor, with a defective rate p_1 , the probability of x defective items appearing among n tested parts is:

$$P(X = x) = \binom{n}{x} p_1^x (1 - p_1)^{n-x} \quad (2)$$

If the batch quality is good, with a defective rate p_2 , the probability of x defective items appearing among n tested parts is:

$$P(X = x) = \binom{n}{x} p_2^x (1 - p_2)^{n-x} \quad (3)$$

Define the likelihood ratio R as:

$$R = \frac{P(X=x|p_1)}{P(X=x|p_2)} \quad (4)$$

The likelihood ratio is used as the test statistic for the sequential sampling model to evaluate the null hypothesis and the alternative hypothesis. If $R > A$, reject the null hypothesis, if $R < B$, accept the null hypothesis, otherwise, continue the inspection.

When R is significantly large, it suggests the sample likely belongs to a batch with poor quality (defective rate p_1). If R is sufficiently large, the batch is deemed poor quality, the product is considered nonconforming, and the batch should be rejected. Conversely, when R is significantly small, the sample likely belongs to a batch with good quality

(defective rate p_2). If R is sufficiently small, the batch is deemed good quality, the product is considered conforming, and the batch should be accepted.

By comparing R with A and B , decisions on acceptance, rejection, or further sampling are made as follows: When $R \geq A$, the batch is considered poor quality and should be rejected. When $R \leq B$, the batch is considered good quality and should be accepted. When $B < R < A$, the batch quality cannot be determined, and further sampling is required until a decision is made.

3.2 A Cost-Benefit Optimization Model for Sampling Based on Dynamic Programming

1) Sampling Cost-Benefit Model

This chapter establishes a sampling cost-benefit model to analyze the six situations encountered by the enterprise, and its goal is to maximize the profit of the enterprise by weighing the five stages of decision-making. Each stage of the decision will directly affect the enterprise's production costs, product quality and market competitiveness, the specific decision-making process in Fig.1.

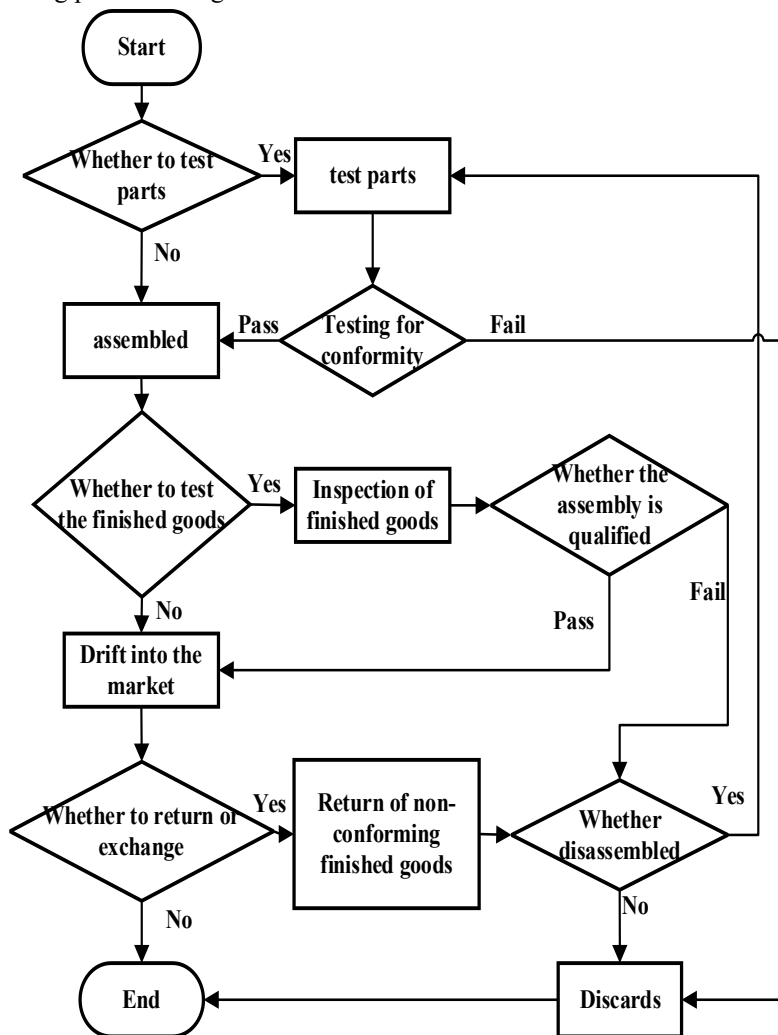


Fig.1. Sampling process

For the complex decision process involved in Figure 1, different control variables must be defined at each key decision point. These variables describe the various inspection and disassembly strategies that determine whether or not to perform an inspection or disassembly operation, so this paper defines variables such as those in Table 3.

Table 3. Decision Variables

Description	Variable	Explanation
Whether to inspect Component 1	x_1	0 for no inspection, 1 for inspection
Whether to inspect Component 2	x_2	0 for no inspection, 1 for inspection
Whether to inspect the final product	x_3	0 for no inspection, 1 for inspection
Whether to disassemble defective products	x_4	0 for no disassembly, 1 for disassembly
Whether to disassemble returned products	x_5	0 for no disassembly, 1 for disassembly

From the enterprise's perspective, the ultimate goal is to maximize total profit. At each decision-making stage, subsequent costs, sales revenue, and the impact of quality control decisions must be considered. Let the total profit be represented as P , which is formulated as follows:

$$P = R_s - C_{c1} - C_{c2} - C_{fp} - C_r \quad (5)$$

Where:

- R_s : Revenue generated from selling final products.
- C_{c1} : Costs related to Component 1.
- C_{c2} : Costs related to Component 2.
- C_{fp} : Costs related to the final product.
- C_r : Costs associated with product returns.
- θ : Total defect rate (the product of defect rates at each decision stage).

The revenue from selling final products is influenced by the decision to inspect the final product and the defect rate, and is expressed as:

$$R_s = N \cdot (1 - \theta_{fp} \cdot x_3) \cdot S \quad (6)$$

Where:

- N : Total production quantity.
- θ_{fp} : Final product defect rate.
- x_3 : Binary variable indicating whether the final product is inspected.
- S : Market price per unit of the final product.

Purchase Cost (C_{p1}):

$$C_{p1} = N \cdot C_{b1} \quad (7)$$

Where C_{b1} represents the purchase cost per unit of Component 1.

Inspection Cost (C_{i1}):

$$C_{i1} = N \cdot x_1 \cdot C_{d1} \quad (8)$$

Where C_{d1} represents the inspection cost per unit of Component 1, and x_1 indicates whether Component 1 is inspected.

Thus, the total cost for Component 1 is:

$$C_{c1} = C_{p1} + C_{i1} \quad (9)$$

Costs Related to Component 2 (C_{c2}): The costs for Component 2 are calculated similarly to those for Component 1, including purchase and inspection costs:

$$C_{c2} = N \cdot C_{b2} + N \cdot x_2 \cdot C_{d2} \quad (10)$$

Where C_{b2} and C_{d2} represent the purchase cost and inspection cost per unit of Component 2, respectively.

Costs Related to the Final Product (C_{fp}): The costs associated with the final product include three components:

Assembly Cost (C_a):

$$C_a = N \cdot C_{fp1} \quad (11)$$

Inspection Cost (C_{fpd}):

$$C_{fpd} = N \cdot x_3 \cdot C_{fp2} \quad (12)$$

Disassembly Cost (C_{fd}):

$$C_{fd} = N \cdot x_4 \cdot C_{fp3} \quad (13)$$

Where C_{fp1} , C_{fp2} , and C_{fp3} represent the assembly, inspection, and disassembly costs per unit, respectively.

Thus, the total cost for the final product is:

$$C_{fp} = C_a + C_{fpd} - R_{fd} \quad (14)$$

Where R_{fd} is the revenue from disassembled components.

Costs Related to Returns (C_r): If defective final products are sold, they may lead to returns. The associated costs include:

Loss Cost (C_{rl}):

$$C_{rl} = N \cdot \theta \cdot x_5 \cdot (C_{b1} + C_{b2} + C_{fp1}) \quad (15)$$

Replacement Cost (C_{rr}):

$$C_{rr} = N \cdot \theta \cdot x_5 \cdot C_{fp3} \quad (16)$$

Thus, the total cost for returns is:

$$C_r = C_{rl} + C_{rr} - R_{rd} \quad (17)$$

Where R_{rd} is the revenue from disassembled returned products.

Total Defect Rate (θ): The total defect rate is determined by the defect rates of Components 1 and 2, as well as the final product. It is expressed as:

$$\theta = \theta_{c1} \cdot (1 - x_1) + \theta_{c2} \cdot (1 - x_2) + \theta_{fp} \cdot (1 - x_3) \quad (18)$$

Binary Constraints:

$$x_1, x_2, x_3, x_4, x_5 \in \{0, 1\} \quad (19)$$

Logical Constraints: If the final product is not inspected ($x_3 = 0$), no disassembly decision for defective products can be made ($x_4 = 0$). If defective products are not disassembled ($x_4 = 0$), returns cannot be disassembled ($x_5 = 0$).

2) Dynamic Programming

Dynamic programming is a powerful optimization technique designed to solve problems that can be divided into overlapping subproblems with optimal substructures. The method works by systematically exploring all possible states of the problem and solving each state iteratively or recursively, while storing intermediate results to avoid redundant computations. The core steps of dynamic programming typically include defining the problem in terms of states and transitions, formulating recursive relations to express the relationships between subproblems, initializing the base case, and iteratively solving the subproblems to compute the final solution. Dynamic programming is usually divided into two main approaches: top-down and bottom-up. It has been successfully applied in areas such as optimization problems, combinatorial algorithms and real-time decision-making systems. In this paper, the dynamic programming approach is used to solve the sampling cost-benefit model, which makes it possible to maximize the benefits of corporate sampling decisions.

3) Monte Carlo simulation

Monte Carlo simulation is a probabilistic model that is numerically computed through random sampling and statistical methods. Its core idea is to use a large number of random samples to simulate the possible outcomes of a problem, and then infer the statistical properties of the system or process, which is suitable for those complex problems that cannot be solved directly by analytical methods. Monte Carlo simulation is widely used in quality control and production processes for part sampling to assess the quality of production lots and to provide a probabilistic basis for decision making. While traditional sampling methods rely on certain sampling rules to decide whether to accept or reject an entire lot, Monte Carlo simulation provides a more flexible and accurate assessment method by modeling the results of different sampling strategies. The basic steps of Monte Carlo simulation are shown in Table 4.

Table 4. The basic steps of Monte Carlo simulation

Monte Carlo simulation
Step1: A confidence value of 95% is set for the percentage threshold of qualified parts based on the defective rate data for each lot of parts and the sampling cost-benefit scheme.
Step2: Define a probabilistic model that generates a random sampling sample and repeats the sampling for optimization
Step3: By recording the results of each sampling, form a sample set and analyze the proportion of parts that pass and fail in each simulation

4. Results

4.1 Sequential Sampling Model Based on Parts Data of Electronic Enterprises

In this section, based on a nominal value of 0.1 for the substandard rate, a plot of the change in the number of samples with the change in the cumulative likelihood ratio for acceptance at the 90% confidence level versus rejection at the 95% confidence level was calculated as in Fig.2.

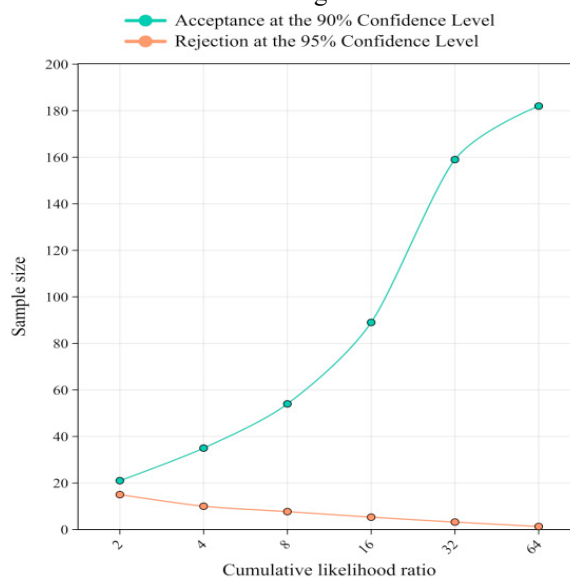


Fig.2. Cumulative likelihood ratio analysis

Through the analysis, it can be seen that the cumulative likelihood ratio in the vicinity of 80 to extract the sample size to meet the requirements of the enterprise decision-making, and this position of the extraction cost is optimal.

4.2 A Cost-Benefit Optimization Model for Sampling Based on Dynamic Programming

This chapter set the maximum number of simulations for 1000, the number of iterations for 100, using MATLAB for Monte Carlo simulation extraction, and the optimal amount of extraction into the dynamic programming algorithm to solve, get the algorithm iterative diagram as in Figure 3 with the optimal decision-making benefit table of the enterprise 6 cases as in Table 4.

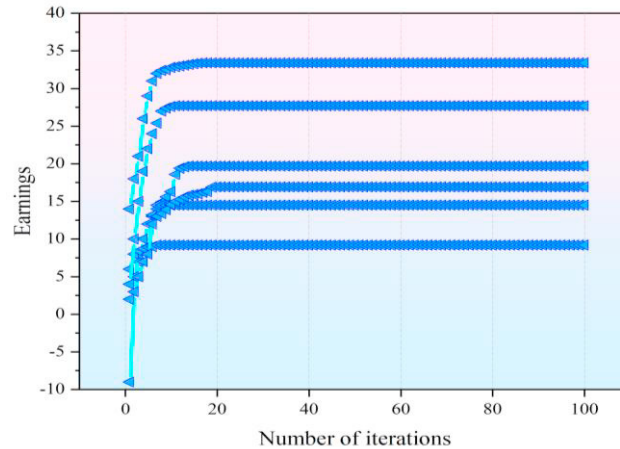


Fig. 3. Algorithm Iteration Diagram

Table 5. The Optimal Decision Making and Benefits of Sampling 6 Scenarios for Firms

Bear fruit	1	2	3	4	5	6
decision	00001	00100	11001	11001	01001	0000
Earnings	23.93	33.53	8.12	14.82	18.67	17.53

5. Conclusion

In this paper, firstly, the model of sequential sampling inspection method is established according to the data of defective rate and selling price of parts and costs provided by the enterprise, secondly, the model of sampling cost benefit is established according to the different sampling costs and other variables, and based on this, the dynamic programming method is introduced to optimize the solution of the sampling cost benefit, and then, the Monte Carlo simulation is used to simulate the sampling, and finally, the optimal sampling decision of the enterprise is concluded and the optimal revenue. The method uses the dynamic programming method to optimize the revenue of the enterprise production of finished products with the variable of whether to sample or not, and adopts the Monte Carlo simulation method to solve the problem of large-scale sampling which is difficult to determine the optimal number of sampling times. The method proposed in this paper overcomes the high cost difficulties of large-scale sampling in reality and provides an effective method for large-scale sampling in enterprise production.

References

- [1] Miroslav V .Reliability analysis of discrete-state performance functions via adaptive sequential sampling with detection of failure surfaces[J].Computer Methods in Applied Mechanics and Engineering,2022,401(PB):
- [2] Scagliarini ,Michele.A sequential test and a sequential sampling plan based on the process capability index Cpmk[J].Computational Statistics,2021,37(3):1-28.
- [3] Houser M T .Explore-exploit behavior in humans as a sequential sampling process[J].Current Psychology,2025,(prepublish):1-13.
- [4] Zylberberg A ,Bakkour A ,Shohamy D , et al.Value construction through sequential sampling explains serial dependencies in decision making.[J].eLife,2024,13

- [5] Toninato G A ,Burkness C E ,Hutchison D W .Spatial distribution and fixed-precision sequential sampling plans for *Popillia japonica* (Coleoptera: Scarabaeidae) adults in primocane raspberry: influence of foliar insecticides[J].Frontiers in Insect Science,2024,41465829-1465829.
- [6] Fomichev M V .On the Complexity of the SequentialSampling Method[J].Journal of Applied and Industrial Mathematics,2024,18(2):227-233.
- [7] Margubur R ,Avijit R ,Nanigopal K , et al.Prevalence and predictors of current tobacco exposure among older adults with chronic disease in India: evidence from large-scale sample survey[J].Journal of Substance Use,2024,29(2):223-231.
- [8] Fan M ,Li Y ,Hu Z , et al.Oropharyngeal swab sampling for PRRSV detection in large-scale pig farms: a convenient and reliable method for mass sampling[J].Porcine Health Management,2024,10(1):39-39.
- [9] Bloise G ,Van L C ,Vailakis Y .An approximation approach to dynamic programming with unbounded returns[J].Journal of Mathematical Economics,2024,111102954-.
- [10] Priorelli M ,Stoianov P I .Dynamic planning in hierarchical active inference[J].Neural Networks,2025,185107075-107075.
- [11] Lu N ,Pan L ,Cui G , et al.The energy management strategy of two-by-one combined cycle gas turbine based on dynamic programming[J].Energy,2024,313134083-134083.
- [12] Psarommatis F ,May G .A Cost - Benefit Model for Sustainable Product Reuse and Repurposing in Circular Remanufacturing[J].Sustainability,2025,17(1):245-245.
- [13] Lovasz Z ,Atmaca Ö ,Veil C , et al.Numerical Analysis of Geometric Influences in Tetrapolar Electrical Impedance Spectroscopy Using Monte Carlo Simulations[J].Current Directions in Biomedical Engineering,2024,10(4):432-435.
- [14] Dario C F ,Aksel F ,Ole Ø , et al.Investigations of the long-term extreme buffeting response of long-span bridges using importance sampling Monte Carlo simulations[J].Engineering Structures,2022,273
- [15] Huang T ,Merwade V .Improving Bayesian Model Averaging for Ensemble Flood Modeling Using Multiple Markov Chains Monte Carlo Sampling[J].Water Resources Research,2023,59(10):
- [16] Djoweyda G ,Megdouda T O .Bayesian autoregressive adaptive refined descriptive sampling algorithm in the Monte Carlo simulation[J].Statistical Theory and Related Fields,2023,7(3):177-187.